

B.Tech. Degree VI Semester Examination April 2018

ME 15-1603 OPERATIONS MANAGEMENT (2015 Scheme)

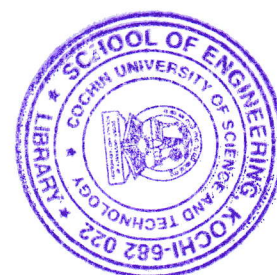
Time: 3 Hours

Maximum Marks: 60

PART A (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define operations management and list its functions.
 (b) Distinguish between PERT and CPM.
 (c) Differentiate between prediction and forecasting.
 (d) Define loading, scheduling and expediting in PPC.
 (e) Write notes on A.B.C and VED methods of inventory control.
 (f) Explain bin cards with its layout.
 (g) Explain Gantt charts with a diagram.
 (h) What are the different priority rules in scheduling?
 (i) List the factors to be considered for locating a factory.
 (j) Define the term depreciation and list the methods of calculating depreciation.



PART B

(4 × 10 = 40)

- II. Draw the network for the following data, find the critical path and project duration. Also find the total float, for the activities. (10)

Activity	A	B	C	D	E	F	G	H	I	J	K	L	M	N
Predecessor	-	A	A	C	C	B, G	D	D	D	F, H	I	E, K	J	L
Time-wks	10	6	2	12	9	4	8	10	5	5	0	7	10	8

OR

- III. (a) Define the term crashing of network, explain with a simple example of your choice. (5)
 (b) Explain the exponential smoothing method of forecasting with an example. (5)
- IV. (a) Define the terms routing and scheduling. Discuss them for (i) continuous (ii) assembly and (iii) job order industry. (5)
 (b) Discuss the procedure you would follow for planning and control of production for a steel furniture industry when orders are received from customers. (5)

OR

- V. (a) Derive the expression for EOQ for basic inventory model and list the assumption used. (5)
 (b) In a manufacturing firm the annual requirement is 24000 units, the supply is instantaneous and shortages are allowed. Each ordering costs ₹350, the inventory holding cost is ₹0.10 per unit per month and the cost of shortage is ₹0.20 per unit per month. Determine the EOQ and the total cost of inventory for this quantity. (5)

(P.T.O.)

- VI. Six jobs are to be scheduled in three machines. The processing time is as follows. Find the optimal schedule so that total elapsed time is minimized. Draw the Gantt chart for the three machines. (10)

Job	J1	J2	J3	J4	J5	J6
M/c1	10	3	5	4	2	1
M/c2	2	4	6	3	1	2
M/c3	8	6	7	9	7	7

OR

- VII. Five jobs arrive at processing station one after another. The processing time and due date are as follows. Use sequencing rules to find (i) total flow time (ii) mean flow time and lateness of job and tell which method is best among. (10)

Job in the sequence of arrival	J1	J2	J3	J4	J5
Processing time days	3	4	2	6	1
Due date (days hence)	5	6	7	9	2

- VIII. (a) Explain the selection factors to be considered for locating a plant. (5)
 (b) What are the most common approaches for plant layout? Compare them. (5)

OR

- IX. (a) What factors influence the selection of material handling equipment? State the application of the following. (5)
 (i) Fork lift truck (ii) Belt conveyor (iii) Jib crane (iv) Roller conveyor.
 (b) The cost of an equipment is ₹7200 and the scrap value is ₹200, the maintenance cost are as follows. When the equipment should be replaced? (5)

Year	1	2	3	4	5	6	7	8
Annual maintenance cost	150	300	450	650	950	1300	1850	2500
